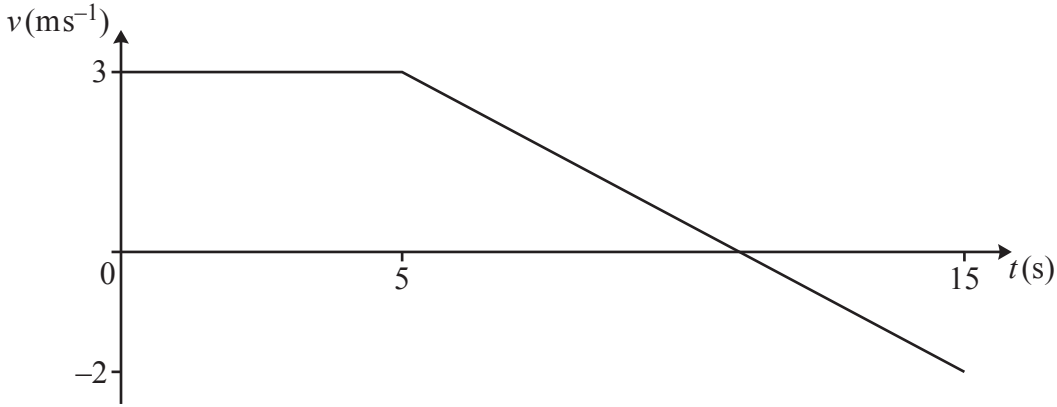
- The acceleration due to gravity is denoted by $g\,\mathrm{m\,s^{-2}}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

6 A car is travelling along a straight horizontal road. The car's velocity $v\,\mathrm{m\,s^{-1}}$ at time $t\,\mathrm{s}$ is shown in the velocity-time graph below. The points (0, 3) and (5, 3) are joined with a line segment, and the points (5, 3) and (15, −2) are joined with another line segment.



(a) Calculate the total distance travelled by the car in the first 15 s. **[3]**

The car is then attached to a caravan by means of a light inextensible horizontal tow bar and continues travelling along the road. You are given the following information.

- The car and caravan accelerate at $1.5\,\mathrm{m\,s^{-2}}$.
- The mass of the car is 1400 kg and the mass of the caravan is 900 kg.
- The driving force acting on the car is D N and the tension in the tow bar is T N.
- The resistances to motion acting on the car and caravan are 400 N and 450 N respectively.

(b) Write down the equations of motion for the car and the caravan separately. **[2]**

(c) Calculate the values of D and T. **[3]**